TEXORA · LECTURE SHEET

Objective Evaluation: FAST, Kawabata and Statistical Analysis

Handle was once judged only by hand and eye. The Kawabata (KES-F) and FAST systems turned handle, drape and tailorability into measured numbers, and statistics turns those numbers into decisions. This sheet covers both instrument systems, formability and control charts, and the mean, standard deviation, CV% and confidence interval every quality report depends on.

TE-3201 · QUALITY EVALUATION OF FABRIC AND APPAREL · TOPIC 9 · LEVEL 3 TERM 2

1. Why handle went objective

For centuries fabric quality was assessed by *handle* (hand): an expert squeezed, bent and stroked the cloth and pronounced it firm or soft, lively or limp. Expert judgement is real skill, but it cannot be written into a contract, telephoned to another continent or plotted on a control chart. Two fabrics agreed as “good handle” in Dhaka could be rejected in London, with no numbers to argue from. Objective evaluation replaces adjectives with low-stress mechanical and surface measurements — the small forces a hand actually applies — so that handle, appearance retention and, above all, *tailorability* (whether a fabric will sew into a smooth, pucker-free garment) can be predicted, specified and traded internationally. Two systems dominate: the research-grade **Kawabata Evaluation System (KES-F)** from Japan and the industry-grade **FAST** system (Fabric Assurance by Simple Testing) developed by CSIRO in Australia.

2. The Kawabata system — KES-F

KES-F measures a fabric in four sensitive modules under very low loads, typically on 20 cm × 20 cm specimens conditioned at 65 % RH, 20 °C. Sixteen parameters are extracted from the loading–unloading curves — the hysteresis (loop) between loading and recovery is itself information, because a hand feels recovery as much as resistance.

MODULE	PARAMETERS	WHAT THEY MEAN
Tensile	EM, LT, WT, RT	EM = extensibility (% extension at maximum load); LT = linearity of the load–extension curve; WT = tensile energy (work to extend); RT = tensile resilience (% energy recovered on unloading)

Bending	B, 2HB	B = bending rigidity per unit width; 2HB = bending hysteresis (moment width at a set curvature) — the liveliness or deadness in bending
Shear	G, 2HG, 2HG5	G = shear rigidity; 2HG and 2HG5 = shear hysteresis at 0.5° and 5° shear angle — governs drape and how a fabric conforms over curves
Compression	LC, WC, RC	LC = linearity of compression curve; WC = compressional energy (softness / fullness of touch); RC = compressional resilience
Surface	MIU, MMD, SMD	MIU = mean coefficient of friction; MMD = mean deviation of friction (stick–slip roughness in friction); SMD = geometrical surface roughness (μm)

2.1 Primary hand values and total hand value (THV)

Kawabata and Niwa connected the sixteen parameters, by regression against large panels of expert judges, to **primary hand values** — scored 0–10 — of which the classic three for suiting are: **Koshi** (stiffness — springy fullness and firmness), **Shari** (crispness — the cool, crackling surface feel of summer fabrics) and **Fukurami** (fullness and softness — warm, bulky loft). Other primary hands (for example smoothness and anti-drape stiffness) suit other fabric classes. The primary hands are then combined, for a stated end use, into a **Total Hand Value (THV)** on a 1–5 scale (5 = excellent, 1 = unacceptable) for that purpose. The same fabric can score THV 5 as a summer shirt and THV 2 as a winter jacket — THV is a fitness-for-purpose judgement computed from physics, not an absolute grade. KES-F remains the reference system for research and for calibrating hand equations; its cost, specimen size and test time kept it out of most factory laboratories, which is the gap FAST was built to fill.

3. The FAST system

FAST is a set of four small, robust instruments plus a set of standard test procedures, designed for mills and garment factories. It measures the properties that decide whether a fabric can be cut, sewn, fused and pressed into a garment that stays smooth — and presents them as a **fingerprint chart** against control limits agreed between fabric supplier and garment maker.

INSTRUMENT	MEASUREMENT	OUTPUT
FAST-1 — Compression meter	Thickness at two low pressures (commonly 0.196 kPa and 9.81 kPa)	Thickness and *surface thickness* (difference between the two) — the compressible surface layer
FAST-2 — Bending meter	Cantilever bending length in warp and weft	Bending rigidity B (converted from bending length and mass per unit area)

FAST-3 — Extension meter	Extension at low loads of 5, 20 and 100 gf/cm, warp, weft and bias	E5, E20, E100 (%); bias extension gives shear rigidity G by the FAST relation
FAST-4 — Dimensional stability test	Hygral expansion (wet–dry size change) and relaxation shrinkage through a wash–dry cycle	Relaxation shrinkage RS and hygral expansion HE (%) in warp and weft

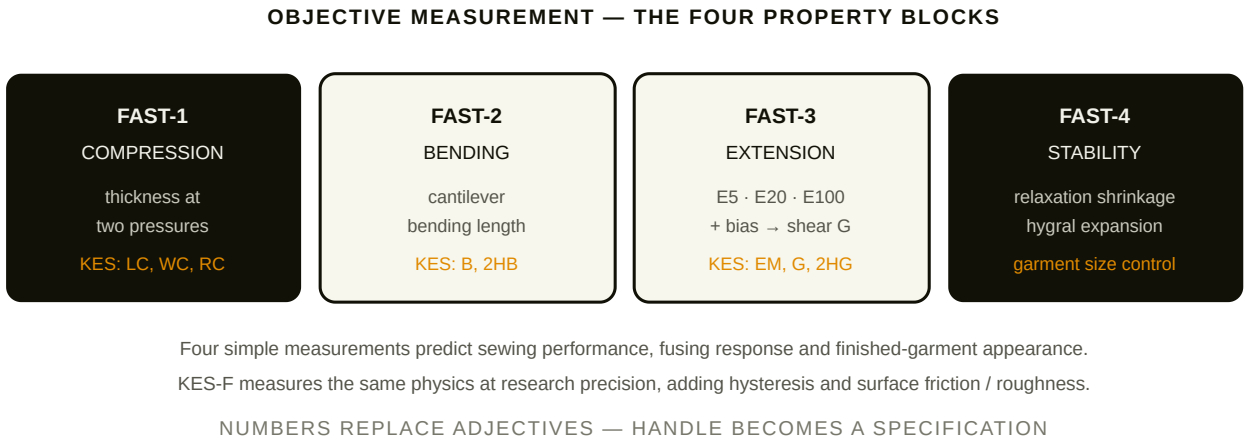


Figure 1 — The FAST instrument set as four property blocks, with the corresponding KES-F parameter groups. Both systems measure low-stress behaviour — the forces a hand and a sewing machine actually apply.

3.1 Formability — the tailoring predictor

The single most used FAST calculation is **formability**: the ability of a fabric to be moulded into a three-dimensional shape (over a shoulder, a chest, a lapel roll) without buckling into wrinkles. A fabric formed beyond its formability limit puckers at the seams — the classic tailoring defect that no pressing can remove. In the SiroFAST definition:

$$\text{Formability } F = (E20 - E5) \times B / 14.7$$

where E20 and E5 are FAST-3 extensions (%) at 20 and 5 gf/cm in the direction considered, B is the FAST-2 bending rigidity (μN·m) in the same direction, and F is obtained in mm^2. The logic is physical: (E20 − E5) is the usable in-plane compressibility the fabric can absorb before buckling, and B is its resistance to buckling — their product is the moulding capacity. Low-formability fabrics (very stable, stiff cloths) need smaller panels, more seams and shaped cutting; high-formability fabrics tailor easily but may grow or sag if stability (FAST-4) is poor. Formability is checked in warp and weft separately, and the lower direction usually controls the garment design.

WORKED EXAMPLE — FORMABILITY

> Warp direction: E5 = 4.0 %, E20 = 12.5 %, bending rigidity B = 55 μN·m.
>
> F = (12.5 − 4.0) × 55 / 14.7 = 8.5 × 55 / 14.7 = 467.5 / 14.7 = **31.8 mm^2**.
>
> Reading: a comfortably formable suiting value — the fabric should mould over shaped seams without pucker, provided its FAST-4 stability is also inside the chart limits. Formability alone never passes a fabric; the fingerprint is judged as a whole.

3.2 FAST control charts (fingerprints) for tailoring

FAST results are plotted on a control chart — the fabric *fingerprint* — with a shaded acceptable zone for each property, set from experience for the garment type. A fabric whose bars all fall inside their zones is predicted to tailor without difficulty. Excursions diagnose the remedy before a single garment is cut: excessive relaxation shrinkage warns of garment size loss (allow in the pattern, or reject); high hygral expansion predicts seam pucker in humid wear and pressing; very low extensibility with low formability predicts pucker at long seams; very high surface thickness predicts fusing and pressing problems. The chart turns an argument about “handle” into a joint supplier–maker decision with the evidence on one page — this, more than the instruments, is why FAST spread through the suiting trade.

4. Statistics for quality-control data

Every test in this course produces a small set of numbers that varies from specimen to specimen. Statistics is how those numbers become a trustworthy mean, an honest spread and a defensible pass/fail decision. Four tools cover most textile QC work.

$$\text{Mean:}\quad \bar{x} = \Sigma x / n \qquad \text{Sample SD:}\quad s = \sqrt{[\ \Sigma(x - \bar{x})^2 / (n - 1)\ ]}$$

$$\text{CV \%} = (s / \bar{x}) \times 100 \qquad \text{95 \% CI of the mean} \approx \bar{x} \pm t \times s / \sqrt{n}$$

- **Mean** — the best estimate of the lot value; specifications on means are judged against it (see Topic 10 for mean vs individual-value specifications).
- **Standard deviation (SD, s)** — the typical scatter of individual results, in the same units as the measurement. The sample SD divides by (n − 1).
- **Coefficient of variation (CV %)** — SD as a percentage of the mean; the unit-free uniformity figure that lets a GSM CV be compared with a strength CV.
- **Confidence interval (CI)** — the range that, with stated confidence (usually 95 %), contains the true lot mean. Its half-width shrinks with √n: quadrupling the number of tests halves the uncertainty. The multiplier t comes from the t-distribution for (n − 1) degrees of freedom (for n = 10, t ≈ 2.26 at 95 %).

WORKED EXAMPLE — GSM STATISTICS (MUST BE REPRODUCIBLE FROM THE RAW DATA)

Ten fabric specimens give GSM readings (g/m^2): 182, 186, 184, 188, 183, 185, 187, 184, 186, 185.

Sum = 1,850, so mean $\bar{x}$ = 1,850 / 10 = **185.0 g/m^2**.

Deviations from 185: −3, +1, −1, +3, −2, 0, +2, −1, +1, 0. Squares: 9, 1, 1, 9, 4, 0, 4, 1, 1, 0 — sum Σ(deviations^2) = **30**.

Sample variance = 30 / (10 − 1) = 30 / 9 = **3.33**; SD s = √3.33 = **1.83 g/m^2**.

CV = 1.83 / 185.0 × 100 = **0.99 %** — a uniform fabric; GSM CV near 1 % is good mill control.

95 % CI: half-width = 2.26 × 1.83 / √10 = 2.26 × 1.83 / 3.162 = 1.31 ≈ 1.3, so CI = **185 ± 1.3 g/m^2** (183.7 to 186.3 g/m^2). Report: “GSM 185.0 ± 1.3 g/m^2 (95 % CI, n = 10), SD 1.83, CV 0.99 %.”

Against a specification of 185 ± 5 g/m^2 on the mean, the lot passes with confidence — the whole interval sits inside the tolerance.

The discipline is: never report a mean without n, SD or CV, and the test conditions. A mean from 3 specimens and a mean from 30 specimens look identical on paper and are not the same evidence. When a result sits near a specification limit, the CI — not the bare mean — decides whether more testing can resolve the question or the lot is genuinely borderline.

Glossary

KES-F — Kawabata Evaluation System: research-grade low-stress measurement in tensile, bending, shear, compression and surface modules. **Primary hand value** — computed 0–10 score for a hand quality such as Koshi, Shari or Fukurami. **THV** — total hand value, 1–5, for a stated end use. **FAST** — Fabric Assurance by Simple Testing: four factory instruments for compression, bending, extension and dimensional stability. **Formability** — capacity to mould in three dimensions without buckling; F = (E20 − E5) × B / 14.7. **Fingerprint chart** — FAST control chart with acceptable zones per property. **CV %** — standard deviation as a percentage of the mean. **Confidence interval** — range containing the true mean with stated confidence.

Key points

• KES-F measures 16 low-stress parameters including hysteresis; primary hands (Koshi, Shari, Fukurami) combine into a THV for a stated end use.

• FAST measures what tailoring needs: FAST-1 compression, FAST-2 bending, FAST-3 extension (E5, E20, E100), FAST-4 dimensional stability.

• Formability F = (E20 − E5) × B / 14.7 predicts seam pucker before cutting.

• FAST fingerprints judge the whole pattern of properties against control zones — no single number passes a fabric.

• Report every mean with n, SD, CV% and, near a limit, its 95 % CI: 185 ± 1.3 g/m^2 is a decision; 185 alone is not.

Practice

1. Five strength readings (N): 210, 218, 205, 215, 212. Calculate the mean, the sample SD (show Σ(deviations^2)) and the CV%.
2. FAST warp data: E5 = 3.2 %, E20 = 9.8 %, B = 68 μN·m. Calculate the formability and state, with a reason, whether it is more or less formable than the worked example fabric (31.8 mm^2).
3. Explain why KES-F records hysteresis (for example 2HB and RT) when a hand judges a fabric, rather than only the loading stiffness.
4. A GSM mean is 185.0 g/m^2 with SD 1.83 from n = 10 (95 % CI ± 1.3). The buyer's limit for the mean is 184 g/m^2 minimum. A colleague says "185 > 184, definite pass." Correct the reasoning using the confidence interval.

Answer key

1. Sum = 210 + 218 + 205 + 215 + 212 = 1,060; mean = 1,060 / 5 = **212 N**. Deviations: −2, +6, −7, +3, 0; squares: 4, 36, 49, 9, 0; Σ = **98**. Variance = 98 / 4 = 24.5; SD = √24.5 = **4.95 N**. CV = 4.95 / 212 × 100 = **2.33 %**.
2. F = (9.8 − 3.2) × 68 / 14.7 = 6.6 × 68 / 14.7 = 448.8 / 14.7 = **30.5 mm^2**. Slightly *less* formable than 31.8 mm^2: its higher bending rigidity does not quite compensate for its smaller usable extension range (E20 − E5 = 6.6 % vs 8.5 %).
3. A hand does not only load a fabric; it releases it and feels the recovery — springiness, liveliness and whether the fabric feels dead or resilient. Hysteresis parameters quantify exactly that loss: RT is the percentage of tensile energy returned, 2HB the bending moment lost in the loop. Two fabrics with identical stiffness B but different 2HB feel different (one lively, one boardy), and they tailor differently, so loading stiffness alone cannot predict hand.
4. The bare mean exceeds 184, but the 95 % CI is 183.7 to 186.3 g/m^2, which *includes* values below 184 — the true lot mean cannot be asserted to be above the limit at 95 % confidence from these 10 tests. The correct statement is "mean passes, confidence interval straddles the limit — borderline." The remedy is more specimens: quadrupling n to about 40 would roughly halve the half-width to ± 0.65 and, if the mean holds, resolve the decision.